

Finite-Difference Transfer Resolves the Jury Fractional-Power Question

Automated Mathematics Run `fa_banach_001`, lane 2

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Status. Likely-valid `literature_implied_answer` with full scope. The decisive external input is the classical FitzGerald–Horn theorem.

1 Source question and supporting theorem

Mashreghi, Nasri, and Vishwakarma ask Question 8.2 on page 36 of arXiv:2512.24575 [1]. For $I = (0, \rho)$, $N \geq 3$, a nonintegral $\alpha > N - 2$, and $A = (a_{mn}) \in \mathbb{P}_N(I)$, put

$$F_A(x, y) = \sum_{m, n=0}^{N-1} a_{mn} x^m y^n$$

and

$$\mathcal{B}(\alpha, A)_{ij} = \frac{\partial^{i+j}}{\partial x^i \partial y^j} F_A(x, y)^\alpha \Big|_{x=y=0}, \quad 0 \leq i, j \leq N-1. \quad (1)$$

They ask whether $\mathcal{B}(\alpha, A)$ is always positive semidefinite. Their Proposition 8.3 shows that an affirmative answer completes the open case in their Theorem E(b).

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Recall Theorem E(b): it presents with an open case of $N \geq 3$. Looking at the simpler though “convoluted” proof for the $N = 2$ case, it may not be straightforward to follow a similar quantitative path to show that the required determinants are nonnegative. In light of this, we propose a possible simplification and formulate a (future) question via an entrywise transform and a differential operator.

Question 8.2. Suppose $I = (0, \rho)$ for $0 < \rho \leq \infty$, and fix an integer $N \geq 3$. For $\alpha > N - 2$ not an integer, define

$$(8.1) \quad \mathcal{B}(\alpha, A)_{ij} := \frac{\partial^{i+j}}{\partial x^i \partial y^j} \left(a_{00} + \sum_{(m,n) \in \mathcal{K}_{N,N}^*} a_{m,n} x^m y^n \right)^\alpha \Big|_{\substack{x=0 \\ y=0}}.$$

Is it true that $\mathcal{B}(\alpha, A) := (\mathcal{B}(\alpha, A)_{ij})_{i,j=0}^{N-1} \in \mathbb{P}_N$, for all $A \in \mathbb{P}_N(I)$?

The resolution of Question 8.2 yields the open case in Theorem E(b) via the following qualitative formulation of the transform.

Proposition 8.3. Consider the premise of Question 8.2, and suppose $f(x) \equiv x^\alpha$. Then for all $A \in \mathbb{P}_N(I)$,

Figure 1: Question 8.2 in the source paper, page 36.

The supporting result is the FitzGerald–Horn theorem, stated as Theorem 2.1 on page 5 of Guillot–Khare–Rajaratnam [3]: if C is an $N \times N$ positive semidefinite matrix with nonnegative entries, then

$$C^{\circ\alpha} := (c_{pq}^\alpha)_{p,q} \succeq 0 \quad \text{for} \quad \alpha \in \mathbb{N} \cup [N-2, \infty). \quad (2)$$

process demonstrate an interesting connection to Loewner convexity. We conclude the paper by discussing related questions and extensions to other power functions in Section 6

2. CHARACTERIZING ENTRYWISE POWERS THAT ARE LOEWNER POSITIVE

The study of Hadamard powers originates in the work of FitzGerald and Horn [8]. We begin our analysis by stating one of their main results that characterizes the Hadamard powers preserving Loewner positivity.

Theorem 2.1 (FitzGerald and Horn, [8] Theorem 2.2). *Suppose $A \in \mathbb{P}_n([0, \infty))$ for some $n \geq 2$. Then $A^{\circ\alpha} \in \mathbb{P}_n$ for all $\alpha \in \mathbb{N} \cup [n-2, \infty)$. If $\alpha \in (0, n-2)$ is not an integer, then there exists $A \in \mathbb{P}_n((0, \infty))$ such that $A^{\circ\alpha} \notin \mathbb{P}_n$. More precisely, Loewner positivity is not preserved for $A = ((1 + \epsilon ij))_{i,j=1}^n$, for all sufficiently small $\epsilon > 0$ with $\alpha \in (0, n-2) \setminus \mathbb{N}$.*

Thus, $\mathcal{H}_{\text{pos}}(n, n) = \mathbb{N} \cup [n-2, \infty)$ for all $2 \leq n \in \mathbb{N}$. Additionally, Hiai [15] showed that the same results as above hold for the critical exponent for the even and odd extensions ϕ_α and ψ_α :

Theorem 2.2 (Hiai, [15] Theorem 5.1). *If $n \geq 2$ and $\alpha \geq n-2$, then $\alpha \in \mathcal{H}_{\text{pos}}^\phi(n, n) \cap \mathcal{H}_{\text{pos}}^\psi(n, n)$.*

Figure 2: Theorem 2.1 in the supporting paper, restating FitzGerald–Horn.

2 Identification and proof idea

The derivative matrix is a limit of congruences of ordinary entrywise transforms. Indeed, F_A is a positive kernel:

$$F_A(x, y) = v(x)^T A v(y), \quad v(x) = (1, x, \dots, x^{N-1})^T.$$

Sample this kernel at $0, h, \dots, (N-1)h$. The sampled matrix is a congruence of A , hence positive. Apply the entrywise transform, then take a second congruence by the forward-difference matrix. As h tends to zero, its entries converge to the mixed derivatives in (1). This yields a general transfer principle and makes (2) apply directly.

3 A general jet-transfer theorem

Theorem 1 (Entrywise-to-Jury jet transfer). *Let $J \subset (0, \infty)$ be open, let $N \geq 1$, and let $f \in C^{2N-2}(J)$. Suppose the entrywise map*

$$C = (c_{pq}) \mapsto f[C] = (f(c_{pq}))$$

maps every $N \times N$ positive semidefinite matrix with entries in J to a positive semidefinite matrix. Let $A = (a_{mn}) \succeq 0$ have entries in an open interval $I \subset (0, \infty)$, and suppose $a_{00} \in J$. Then the jet matrix

$$\mathcal{J}_f(A)_{ij} := \partial_x^i \partial_y^j f(F_A(x, y)) \Big|_{x=y=0}, \quad 0 \leq i, j \leq N-1,$$

is positive semidefinite.

Proof. For $h > 0$, set $t_p = ph$ for $0 \leq p \leq N-1$ and define the Vandermonde matrix

$$V_h = (t_p^m)_{p,m=0}^{N-1}.$$

The sampled kernel matrix is

$$C_h = (F_A(t_p, t_q))_{p,q=0}^{N-1} = V_h A V_h^T \succeq 0. \quad (3)$$

As $h \downarrow 0$, every entry of C_h tends to $a_{00} \in J$; hence for all sufficiently small h , all entries lie in J . By hypothesis, $f[C_h] \succeq 0$.

Let L_h be the lower-triangular forward-difference matrix

$$(L_h)_{ip} = \begin{cases} h^{-i}(-1)^{i-p} \binom{i}{p}, & 0 \leq p \leq i, \\ 0, & p > i. \end{cases}$$

Congruence preserves positivity, so

$$L_h f[C_h] L_h^T \succeq 0. \quad (4)$$

Its (i, j) entry is precisely

$$\frac{1}{h^{i+j}} \sum_{p=0}^i \sum_{q=0}^j (-1)^{i-p+j-q} \binom{i}{p} \binom{j}{q} f(F_A(ph, qh)),$$

the order- (i, j) forward-difference quotient of $f(F_A(x, y))$ at the origin. Since $f \in C^{2N-2}$, this converges to $\mathcal{J}_f(A)_{ij}$ as $h \downarrow 0$. Thus the matrices in (4) converge entrywise to $\mathcal{J}_f(A)$. The finite-dimensional positive semidefinite cone is closed, proving the claim. $\square$

4 Resolution and exact classification

Corollary 2 (Affirmative answer to Question 8.2). *Under the hypotheses of Question 8.2, $\mathcal{B}(\alpha, A) \succeq 0$ for every $A \in \mathbb{P}_N(I)$. Moreover, for $N \geq 3$,*

$$x^\alpha \text{ preserves positivity under the Jury transform on } \mathbb{P}_N(I) \iff \alpha \in \mathbb{Z}_{\geq 0} \cup [N-2, \infty).$$

Proof. Apply Theorem 1 to $f(t) = t^\alpha$ and use FitzGerald–Horn (2). This proves the assertion of Question 8.2. Proposition 8.3 of [1] gives

$$Df_\diamond(A)D = \mathcal{B}(\alpha, A), \quad D = \text{diag}(1, 1, 2!, \dots, (N-1)!),$$

so the Jury transform is positive as well. Nonnegative integral powers are also covered (or follow by repeated Jury products). Conversely, Theorem E(b) of [1] proves failure for every $\alpha < N-2$ that is not a nonnegative integer. This gives the exact classification. $\square$

5 Provenance, verification, and scope

This is an agent-identified implication, not a later paper whose authors explicitly knew they were answering Question 8.2. The decisive theorem is from 1977 and is quoted in the source itself; the new identification is the finite-difference jet limit. Under the run protocol, the conservative status is therefore `literature_implied_answer` (`full resolution`) rather than an original full solution.

The numerical checker computes (1) by a truncated bivariate binomial series. It tested 390,000 random positive-entry Gram matrices for $N = 3, 4$ and eight exponents above the threshold, with ranks from one through N . No violation below a normalized tolerance of -10^{-8} was found. The numerical test is not used in the proof.

A bounded search through 9 August 2026 checked the current arXiv and journal versions of the source, arXiv:1311.1581, the FitzGerald–Horn article metadata, later Jury-product papers arXiv:2602.21056 and 2607.13251, and exact/close formula searches. No source recording this precise implication was found. No other open problems in arXiv:2512.24575 are addressed here.

Human-review recommendation. Check the indices in L_h and confirm that the dimension in FitzGerald–Horn is exactly the number N of sampled points. If those two points pass, Question 8.2 and the fractional-power gap in Theorem E are fully resolved.

References

- [1] J. Mashreghi, M. Nasri, and P. K. Vishwakarma, *Functional calculi, positivity, and convolution of matrices*, arXiv:2512.24575 (2025); Journal of Mathematical Analysis and Applications, Article 130934 (2026).
- [2] C. H. FitzGerald and R. A. Horn, *On fractional Hadamard powers of positive definite matrices*, J. Math. Anal. Appl. **61** (1977), 633–642.
- [3] D. Guillot, A. Khare, and B. Rajaratnam, *Complete characterization of Hadamard powers preserving Loewner positivity, monotonicity, and convexity*, J. Math. Anal. Appl. **425** (2015), 489–507, arXiv:1311.1581.